

Real-World Evidence: What Is Missing, Precisely

Status: the architecture is IMPLEMENTED. The evidence is a FIXTURE. **Blocker:** LIVE_DATA — BLOCKED_EXTERNAL, unchanged. **Executable form:** pkg/casefabric/semantics.go · asserted by semantics_test.go

1. The gap, stated once

VERIQO's six domains now declare their semantics as data: what they work in, what counts as evidence, what their claims must show, what would falsify them, and how a case ends. Thirty-five evidence classes across six domains, each naming a real source.

Every one of those thirty-five is currently a fixture. Not a stub, not a mock — a hand-authored value in a test. The chain executes over them correctly, which proves the machinery composes. It proves nothing about behaviour on real rights-aware commercial data.

That is the largest remaining distance between *engineering excellence* and a *commercial intelligence platform*, and no amount of further engineering closes it. It closes when somebody signs a data agreement.

2. What "rights-aware" means here, and why it is the hard part

The naive reading of this gap is "connect the APIs". It is not that.

Every one of these sources arrives with a licence, and the licence governs what VERIQO may do with the data — not merely whether it may fetch it. The ten purpose-bound rights in pkg/disclosure/access exist because VIEW, EXPORT, AI_PROCESS, RAG and TRAIN are five separate grants, and most commercial data agreements grant some and refuse others.

So a real connector is not an HTTP client. It is:

- a **licence** whose terms are encoded as rights, not summarised in prose;
- an **acquisition authority** check that runs *before* contact (Article 4 — a rights failure denies before the request is made, never after);
- a **provenance record** naming source, licence and acquisition path;
- an **independence assessment** — many of these sources are aggregators, and three feeds off one root are one source (Article 3);
- a **retention and erasure** position, because commercial data agreements routinely require deletion on termination.

pkg/authz, pkg/qualification/independence and pkg/governance/data already implement all five. What is missing is a real licence to run them against.

3. The thirty-five classes, and how much is party-mediated

Fourteen of the thirty-five are **party-mediated**: acquired through a party to the matter rather than independently. That is not a defect — a bill of lading is the carrier's document and there is no independent copy — but it is a fact about what those classes can establish, and Semantics.PartyMediatedClasses() reports it per domain rather than letting a domain describe itself more flatteringly than the evidence allows.

TestEveryDomainHasPartyMediatedEvidence fails for any domain claiming all its evidence is independently acquired, because for a real matter that claim is not credible.

4. Per-source acquisition requirements

Source family	Domains	What acquisition actually requires
AIS (terrestrial + satellite)	maritime	A commercial aggregator agreement. Coverage is uneven and the gaps matter evidentially — an AIS gap is an <i>absence</i> , and carries weight only after the nine-condition observability gate
SAR imagery	maritime	A tasking or archive agreement with an SAR provider. Tasking has lead time, which is a live constraint on what can be evidenced after the fact
Optical EO	maritime	An EO provider agreement. Cloud cover is a coverage-adequacy question, not an inference
Bill of lading / eBL	commodity, trade finance	Carrier or MLETR-conformant platform access. Platform reliability is itself an external qualification question — VERIQO records the platform's assertion and does not certify it
Customs	supply chain	Authority-by-authority access, each with its own legal basis. Not a single integration
Sanctions lists	supply chain	Issuing-authority feeds, versioned : the list in force at the material time governs, not today's list (Article 7)
Insurance / P&I	insurance	Bilateral agreements with insurers and clubs. Most of what arrives is party-mediated by construction
Port call records	maritime	Port authority or agent access, jurisdiction by jurisdiction
Weather hindcast	maritime	A meteorological service agreement; usually the least encumbered of these
Dark web	(not built)	An acquisition capability that does not exist. It is DESIGNED and named as such — see the constitutional proof audit
Aureum	(not built)	Not built

5. What would change when real data arrives — and what would not

Would change. The evidence set stops being hand-authored. Independence assessment starts returning DEPENDENT and UNKNOWN verdicts that no fixture produces. Coverage gaps become real, so the observability gate starts refusing absences that the fixtures let through. Qualification states other than SUPPORTED start appearing, and the reverse-proof gap analysis starts reporting Unattempted requirements nobody can obtain.

That is the point. The fixtures exercise the happy path; real data exercises the refusals, and the refusals are the product.

Would not change. No package. The connectors would populate `pkg/evidence/manifest` exactly as the fixtures do. The chain, the qualification rules, the proof object, the case fabric and the ledger are indifferent to where evidence came from, which is what makes them worth having.

6. Honest status

Architecture for rights-aware acquisition	IMPLEMENTED — authz, independence, provenance, retention
Domain semantics naming real sources	IMPLEMENTED — 35 classes, 6 domains, declared as data
Real data agreements	NONE
LIVE_DATA blocker	BLOCKED_EXTERNAL — unchanged, and this document does not move it

Appendix — the six domains as declared

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=== MARITIME ===
ONTOLOGY      Vessel, Voyage, Port, Event, Causation, Loss
EVIDENCE      AIS position track <- AIS aggregator (terrestrial and satellite)
              SAR imagery <- synthetic-aperture radar provider
              optical EO imagery <- earth-observation provider
              port call record <- port authority or agent
              weather hindcast <- meteorological service
              vessel log extract <- the operator [party-mediated]
OBLIGATIONS   class survey report <- classification society
              the vessel deviated from its declared route
                requires: a declared route; a position track covering the window; an observability verdict for any gap in the track
                falsified by: a position track within the declared corridor for the whole window
              a dark period was deliberate rather than a coverage gap
                requires: coverage adequacy for the window; an OBSERVED_ABSENT verdict, not an unattempted one; a rival explanation to
                falsified by: a coverage gap in the source that explains the absence without any act of the vessel
RULES          an AIS gap is an absence, and an absence carries weight only after the nine-condition observability gate
              vessel identity below the resolution threshold stays unresolved rather than merging two vessels
OUTCOMES       position from a single provider is a single source however many feeds it aggregates
              findings_issued, evidence_package_delivered, no_further_action, referred_to_authority

=== COMMODITY ===
ONTOLOGY      Cargo, Shipment, Document, Loss, Quantum, Causation
EVIDENCE      independent assay certificate <- accredited laboratory
              draft survey report <- appointed surveyor
              loading tally <- the terminal [party-mediated]
              sealed sample chain of custody <- the sampling party [party-mediated]
              temperature and humidity log <- container telemetry provider
              bill of lading <- the carrier [party-mediated]
OBLIGATIONS   the cargo was off-specification before loading
                requires: a pre-load sample analysed by an independent laboratory; an unbroken sample chain of custody; the in-transit
                falsified by: a clean pre-load sample from a properly sealed and custodied specimen
              the shortage arose in carriage rather than at loading
                requires: a load-port figure; a discharge-port figure; a stated measurement tolerance for both
                falsified by: a discrepancy within the combined measurement tolerance of the two surveys
RULES          an assay from a laboratory a party appointed is party-mediated, and independence is assessed accordingly
              quantity claims are never established from a single measurement without a stated tolerance
              a sealed sample with a broken custody chain is evidence of the break, not of the contents
OUTCOMES       quality_determined, evidence_package_delivered, no_further_action, referred_to_arbitration

=== SUPPLYCHAIN ===
ONTOLOGY      Shipment, Organization, Event, Breach, Responsibility, Timeline
EVIDENCE      customs declaration <- the customs authority
              sanctions and screening list <- the issuing authority
              carrier milestone feed <- the carrier [party-mediated]
              supplier audit report <- an appointed auditor
              certificate of origin <- the issuing chamber or authority
              purchase order and contract <- the transacting parties [party-mediated]
OBLIGATIONS   the disruption originated at a named tier-2 supplier
                requires: a traced chain from the disruption to that tier; the intermediate tiers excluded; an alternative origin test
                falsified by: an intermediate tier that independently accounts for the disruption
              the consignment involves a sanctioned party
                requires: the list version in force at the material time; an identity match above the resolution threshold; the beneficiary
                falsified by: a name match that fails identity resolution, which is a coincidence of names
RULES          a sanctions name match is a hypothesis until identity resolution clears the threshold; it is never a finding on its own
              the list version in force at the material time governs, not today's list -- Article 7
              a carrier's own milestone feed is party-mediated and is not independent corroboration of the carrier's performance
OUTCOMES       origin_established, evidence_package_delivered, no_further_action, referred_to_authority

=== INSURANCE ===
ONTOLOGY      Claim, Policy, Loss, Quantum, Causation, Obligation
EVIDENCE      policy wording and endorsements <- the insurer [party-mediated]
              loss adjuster report <- an appointed adjuster
              P&I club correspondence <- the club [party-mediated]
              repair invoice and quotation <- the repairer
              survey report <- an appointed surveyor
              notice of claim <- the claimant [party-mediated]
OBLIGATIONS   the loss falls within the policy's insured perils
                requires: the policy version in force at the loss; the proximate cause established; each exclusion considered and addressed
                falsified by: an exclusion that applies on the established facts
              the quantum is the sum claimed
                requires: an evidence-backed amount for every component; the deductible and limits applied; any betterment identified
                falsified by: a component with no evidential backing, or double-counting between components
RULES          coverage is a question for the insurer and the policy; VERIQ0 establishes facts and does not determine liability
              every amount cites the evidence it rests on; an unbacked figure is not a quantum
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OUTCOMES	payment authority and payment execution authority are disjoint role sets -- one party never does both evidence_package_delivered, quantum_computed, no_further_action, referred_to_arbitration
=== TRADEFINANCE ===	
ONTOLOGY	Contract, Clause, Document, Transaction, Obligation, Breach
EVIDENCE	documentary credit and amendments <- the issuing bank presented document set <- the beneficiary [party-mediated] electronic bill of lading record <- an MLETR-conformant eBL platform inspection certificate <- the named inspection body SWIFT message trace <- the messaging network
OBLIGATIONS	the presentation is compliant with the credit requires: the credit and every amendment in force; each required document present; each discrepancy identified against falsified by: a document that fails a stated term of the credit the eBL holder is the party claiming to be requires: the platform's transfer record; an unbroken transfer chain from issuance; the platform's own reliability ass falsified by: a transfer in the chain that the platform cannot evidence
RULES	discrepancy is determined against a stated term of the credit, never as a general impression of the document set an eBL platform's reliability is an external qualification question; VERIQO records the platform's assertion and does no
OUTCOMES	a document a party presented is party-mediated evidence of the transaction, not independent evidence of the facts it rec determination_issued, evidence_package_delivered, no_further_action, referred_to_issuing_bank
=== DISPUTE ===	
ONTOLOGY	Claim, Counterclaim, Contradiction, ProofObligation, Timeline, Finding
EVIDENCE	pleadings and statements of case <- the parties [party-mediated] disclosed document set <- the disclosing party [party-mediated] expert report <- an appointed expert witness statement <- the witness [party-mediated] contemporaneous record <- whichever party held it [party-mediated]
OBLIGATIONS	the evidence package is complete for the issues framed requires: every framed issue decomposed into supporting, contradicting and missing evidence; each party's position rec falsified by: a framed issue with no evidence decomposition
RULES	VERIQO furnishes facts, evidence, timelines, contradictions, causation hypotheses, quantum and proof obligations; the ar two parties' positions sitting side by side, unreconciled, IS the output -- neither is weighted a legal question is marked AWAITING_LEGAL_INTERPRETATION and left there; the system does not answer it
OUTCOMES	evidence_package_delivered, no_further_action, matter_withdrawn